

OCTOBER 2026

Predictive Mock Test SOLUTION BOOK

ENGINEERING PATHWAY

81

questions

3

modules

40

minutes per module

MODULES IN THIS BOOK

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Original practice material for ESAT preparation.

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MODULE 01

Mathematics 1

QUESTIONS & WORKED SOLUTIONS

Engineering pathway

27

questions

40

minutes

1

mark per correct answer

NO CALCULATOR

Using this module

Choose one answer per question. There is no negative marking.
Each worked solution appears immediately below its question.
Attempt the question first; cover the solution if working to time.
Use separate paper for your working.

Question 1 begins on the next page.

Question 01

A rectangular card has a length recorded as 12 cm and a width recorded as 8 cm. Both measurements are rounded to the nearest centimetre. What is the least upper bound for the ratio of its actual length to its actual width?

A $\frac{3}{2}$

B $\frac{5}{3}$

C $\frac{23}{15}$

D $\frac{25}{17}$

E $\frac{25}{16}$

Worked solution

Answer B

Write the actual length and width as L and W . Their bounds are $11.5 \leq L < 12.5$ and $7.5 \leq W < 8.5$. The largest ratios use a length approaching 12.5 and the smallest width, 7.5. Thus $\frac{L}{W} < \frac{12.5}{7.5} = \frac{5}{3}$, and ratios can approach $\frac{5}{3}$ as closely as desired. The least upper bound is $\frac{5}{3}$.

Trap: using both upper bounds. To make a positive ratio as large as possible, make its numerator large and its denominator small. The upper bound need not be attained.

Question 02

For a real number k , the equation $x^2 - (k + 4)x + 2k = 0$ has one root strictly between 0 and 1, and its other root is greater than 2. What is the set of all possible values of k ?

A $k < 0$

B $0 \leq k \leq 3$

C $k > 3$

D $0 < k < 3$

E $k < 0$ or $k > 3$

Worked solution

Answer D

Let $p(x) = x^2 - (k + 4)x + 2k$. A root in $(0, 1)$ requires $p(0) > 0$ and $p(1) < 0$, giving $k > 0$ and $k < 3$. Also $p(2) = -4 < 0$, so after the first crossing the other root is beyond 2.

Trap: checking only the discriminant and forgetting where the roots lie.

Question 03

What is the value of x that satisfies $\frac{9^{x+1} 27^{2-x}}{3^{x-1}} = 81$?

A $\frac{5}{2}$

B $\frac{3}{2}$

C 2

D $\frac{1}{2}$

E $\frac{7}{2}$

Worked solution

Answer A

Write every number as a power of 3. The equation becomes $3^{2(x+1)+3(2-x)-(x-1)} = 3^4$.
Hence $9 - 2x = 4$, giving $x = \frac{5}{2}$.

Trap: subtract the whole denominator exponent $x - 1$, so its contribution is $-x + 1$.

Question 04

The least positive integer n leaves remainder 2 when divided by 5, and remainder 3 when divided by 7. What is the remainder when $n^2 + 4n + 1$ is divided by 35?

A 3

B 5

C 18

D 13

E 8

Worked solution

Answer E

The candidates leaving remainder 2 on division by 5 are 2, 7, 12, 17, The first that leaves remainder 3 on division by 7 is $n = 17$. Then $n^2 + 4n + 1 = 358 = 10 \cdot 35 + 8$, so the remainder is 8.

Question 05

A container holds 4 red, 3 blue and 2 green counters. Three counters are drawn without replacement. Given that exactly one of the three is green, what is the probability that the first counter drawn is red?

A $\frac{1}{7}$

B $\frac{8}{21}$

C $\frac{2}{9}$

D $\frac{4}{9}$

E $\frac{1}{2}$

Worked solution

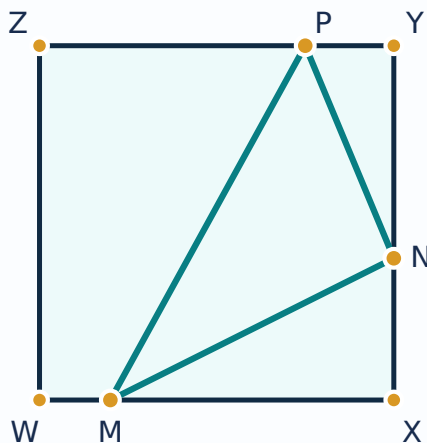
Answer B

Given that exactly one of the three counters is green, the green counter is equally likely to occupy any of the three positions. Therefore the first counter is non-green with probability $\frac{2}{3}$. Conditional on its being non-green, each of the seven non-green counters is equally likely to be first, and four are red. Hence the required probability is $\frac{2}{3} \times \frac{4}{7} = \frac{8}{21}$.

Trap: conditioning changes the sample space; the unconditional probability $\frac{4}{9}$ is not the answer.

Question 06

$WXYZ$ is a square of side 1. Points M, N, P lie on WX, XY, YZ , respectively, with $WM : MX = 1 : 4$, $XN : NY = 2 : 3$, and $YP : PZ = 1 : 3$. What is the area of triangle MNP ?



Schematic diagram; not to scale.

A $\frac{1}{5}$

B $\frac{1}{4}$

C $\frac{29}{100}$

D $\frac{3}{10}$

E $\frac{7}{20}$

Worked solution

Answer C

The ratios give $WM = \frac{1}{5}$, $MX = \frac{4}{5}$, $XN = \frac{2}{5}$, $NY = \frac{3}{5}$, $YP = \frac{1}{4}$, and $PZ = \frac{3}{4}$. Subtract the two right triangles MXN , NYP , and the trapezium $WMPZ$ from the unit square.

The two triangle areas are $\frac{1}{2} \cdot \frac{4}{5} \cdot \frac{2}{5} = \frac{4}{25}$ and $\frac{1}{2} \cdot \frac{3}{5} \cdot \frac{1}{4} = \frac{3}{40}$. The trapezium has parallel sides WM and ZP , separated by distance 1, so its area is $\frac{1}{2}(\frac{1}{5} + \frac{3}{4}) = \frac{19}{40}$. Thus the required area is $1 - \frac{4}{25} - \frac{3}{40} - \frac{19}{40} = \frac{29}{100}$.

Tip: subtract familiar shapes from the square; no general coordinate-area formula is needed.

Question 07

Pumps A and B fill a tank at constant rates. In a trial, increasing A's rate by 20% and decreasing B's rate by 10% leaves their combined rate unchanged.

At its original rate, A alone fills the tank in 30 minutes. The empty tank is now filled using both pumps at their original rates. After 4 minutes, B is switched off and A completes the filling.

What is the total time taken to fill the tank?

A 14 minutes

B 18 minutes

C 20 minutes

D 22 minutes

E 24 minutes

Worked solution

Answer D

Let the original filling rates be a and b tanks per minute. The trial gives

$$1.2a + 0.9b = a + b \Rightarrow 0.2a = 0.1b \Rightarrow b = 2a.$$

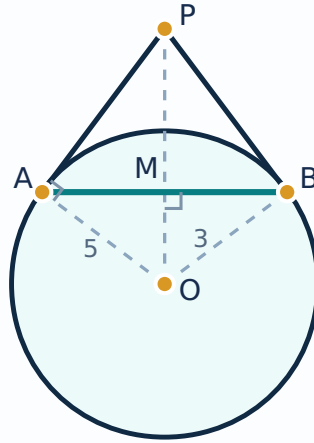
Since $a = 1/30$, the combined rate is $3a = 1/10$ tank per minute. In the first 4 minutes, the pumps fill $4/10 = 2/5$ of the tank.

A fills the remaining $3/5$ in $(3/5) \times 30 = 18$ minutes. The total is $4 + 18 = 22$ minutes.

Useful approach: Use the unchanged combined rate to find the ratio of the two original rates before calculating any times.

Question 08

A circle has centre O and radius 5. Chord AB is 3 units from O , measured perpendicularly. The tangents at A and B meet at P . What is OP ?



Schematic diagram; M is the midpoint of AB .

A $\frac{25}{3}$

B $\frac{25}{6}$

C 5

D 10

E 15

Worked solution

Answer A

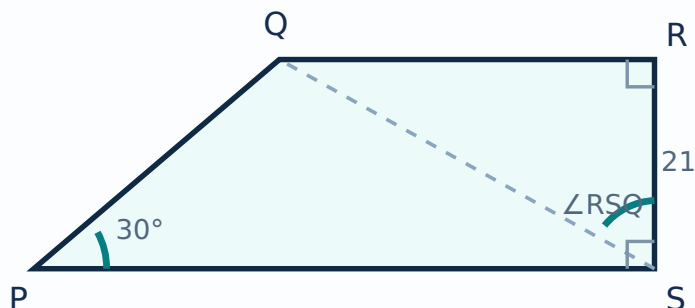
Let M be the midpoint of chord AB . The perpendicular from O to AB passes through M , and O, M, P are collinear by symmetry. We have $OM = 3$ and $OA = 5$.

Triangle OMA is right-angled at M , while triangle OAP is right-angled at A , because a radius is perpendicular to a tangent. They share the angle at O , so they are similar.

Hence $\frac{OM}{OA} = \frac{OA}{OP}$, giving $\frac{3}{5} = \frac{5}{OP}$ and $OP = \frac{25}{3}$.

Question 09

$PQRS$ is a right trapezium with $PS \parallel QR$, right angles at R, S , $RS = 21$, $\angle QPS = 30^\circ$, and $\tan \angle RSQ = \frac{4}{3}$. What is PS ?



Schematic diagram; diagonal SQ is shown for the given angle.

A $28 + 21\sqrt{3}$

B $21 + 21\sqrt{3}$

C $21 + 28\sqrt{3}$

D $49 + \sqrt{3}$

E $28 + 7\sqrt{3}$

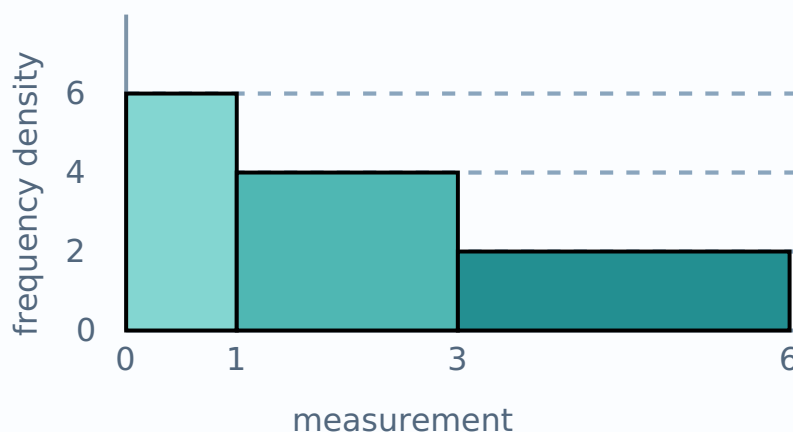
Worked solution

Answer A

In right triangle QRS , $QR/RS = 4/3$, so $QR = 28$. The horizontal distance from P to the vertical through Q is $21/\tan 30^\circ = 21\sqrt{3}$. Therefore $PS = 28 + 21\sqrt{3}$.

Question 10

A histogram represents exactly 20 measurements. Its class intervals are $0 \leq x < 1$, $1 \leq x < 3$, and $3 \leq x \leq 6$, where x is the measurement. The frequency densities are shown below.



The unequal widths are part of the histogram.

The mean of all 20 measurements is exactly 1.8. What is the greatest possible number of measurements that are at least 5?

A 3

B 5

C 8

D 4

E 6

Worked solution

Answer B

Frequency equals class width multiplied by frequency density, so the three frequencies are $1 \times 6 = 6$, $2 \times 4 = 8$, and $3 \times 2 = 6$. The total of all measurements is $20(1.8) = 36$.

Only the six observations in $[3, 6]$ can be at least 5. If all six were, the total would be at least $6(5) + 8(1) + 6(0) = 38$, which is impossible. Five are achievable: take five values 5, one value 3, eight values 1, and six values 0. These respect the stated class boundaries and sum to 36. The maximum number is therefore 5.

Question 11

Two fair six-sided dice are rolled. Given that at least one die shows 4, what is the probability that the sum of the scores is at least 9?

A $\frac{1}{3}$

B $\frac{2}{5}$

C $\frac{4}{11}$

D $\frac{5}{12}$

E $\frac{1}{2}$

Worked solution

Answer C

Given that at least one die shows 4, there are $6 + 6 - 1 = 11$ equally likely ordered outcomes. The outcomes with sum at least 9 are (4, 5), (4, 6), (5, 4), (6, 4). Therefore the probability is $4/11$.

Question 12

For positive integers n , define $a_n = n^2 + 3n + 2$. How many of the first 120 terms are divisible by 12?

A 32

B 36

C 44

D 40

E 48

Worked solution

Answer D

Factor $a_n = (n+1)(n+2)$. Divisibility by 12 requires divisibility by both 3 and 4.

For the factor 4, the even member of the two consecutive integers must be a multiple of 4. Thus n leaves remainder 2 or 3 when divided by 4.

For the factor 3, n must leave remainder 1 or 2 when divided by 3. In each block of 12 consecutive values of n , both requirements hold at remainders 2, 7, 10, 11.

There are $120/12 = 10$ complete blocks, each with 4 successes, giving 40 terms.

Question 13

The graph of $y = g(x)$ is made from straight-line segments joining, in order, the points $(-4, 3)$, $(-1, -3)$, $(2, 1)$, $(5, -2)$, $(7, 2)$. What is the sum of the x -coordinates of all its intersections with the x -axis?

A $\frac{29}{4}$

B $\frac{31}{4}$

C 8

D $\frac{33}{4}$

E $\frac{35}{4}$

Worked solution

Answer B

On the first segment, the heights 3 and -3 put the crossing halfway between $x = -4$ and $x = -1$, at $-\frac{5}{2}$. On the second segment, the height rises from -3 to 1, so zero is reached three quarters of the way from -1 to 2: $-1 + \frac{3}{4}(3) = \frac{5}{4}$.

On the third segment, the height falls from 1 to -2 , so the crossing is one third of the way from 2 to 5, at 3. The final segment crosses halfway from 5 to 7, at 6. Their sum is $-\frac{5}{2} + \frac{5}{4} + 3 + 6 = \frac{31}{4}$.

Question 14

For a real parameter a , consider $x^2 - 2(a+1)x + a^2 - 1 = 0$. For which values of a does this equation have one positive root and one negative root?

A $a \leq -1$

B $-1 \leq a < 0$

C $-1 < a < 1$

D $0 < a \leq 1$

E $a < -1$ or $a > 1$

Worked solution

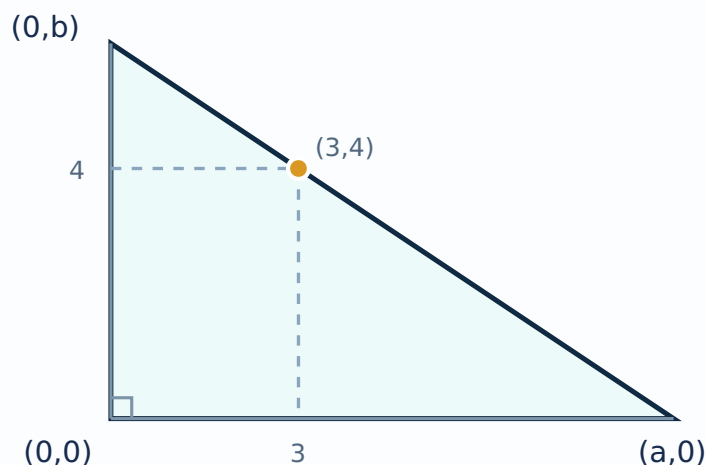
Answer C

Writing a monic quadratic as $(x-r)(x-s)$ shows that its constant term is the product rs . Here the product is $a^2 - 1$. Opposite signs require a negative product, so $a^2 - 1 < 0$, giving $-1 < a < 1$.

This condition is also sufficient: the upward-opening quadratic is negative at $x = 0$ and positive for sufficiently large positive or negative x , so it crosses the axis once on each side of zero. The endpoints $a = \pm 1$ are excluded because one root is then zero.

Question 15

A right triangle has vertices $(0,0)$, $(a,0)$, $(0,b)$, where $a, b > 0$. The point $(3,4)$ lies on its hypotenuse. What is the minimum possible area of the triangle?



Schematic diagram; not to scale.

A 18

B 20

C 32

D 28

E 24

Worked solution

Answer E

The line through $(a,0)$ and $(0,b)$ has equation $y = b - \frac{b}{a}x$, equivalently $\frac{x}{a} + \frac{y}{b} = 1$. Since $(3,4)$ lies on it, $\frac{3}{a} + \frac{4}{b} = 1$. Set $u = \frac{3}{a}$ and $v = \frac{4}{b}$, so $u + v = 1$ and $ab = \frac{12}{uv}$.

Because $(u - v)^2 \geq 0$, we have $4uv \leq (u + v)^2 = 1$. Therefore $ab \geq 48$, and the area is at least $\frac{1}{2}(48) = 24$. Equality occurs at $u = v = \frac{1}{2}$, giving $a = 6$ and $b = 8$, so this minimum is attained.

Question 16

A sequence is defined by $a_1 = 1$ and $a_{n+1} = 3a_n + 2$. What is the remainder when a_{2026} is divided by 7?

A 4

B 2

C 1

D 5

E 6

Worked solution

Answer A

The successive remainders on division by 7 are 1, 5, 3, 4, 0, 2, then repeat. Since $2026 = 6 \cdot 337 + 4$, the required remainder is the fourth one in the cycle, namely 4.

Question 17

Points $A = (-4, 7)$, $B = (5, -2)$ and $C = (3, 6)$ are given. Point P lies on AB such that $AP : PB = 2 : 1$. Point Q is the reflection of C in P , meaning a rotation of 180° about P . What is $2x_Q - y_Q$?

A -6

B -2

C 2

D 6

E 10

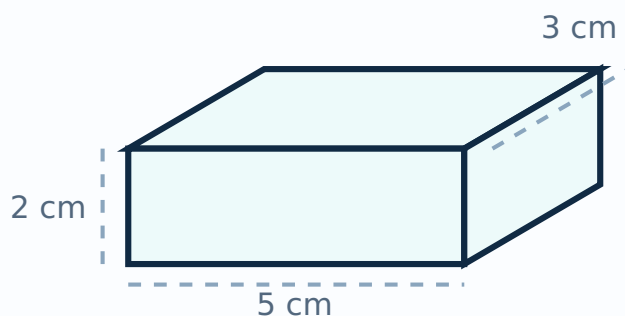
Worked solution

Answer D

$P = A + \frac{2}{3}(B - A) = (-4, 7) + \frac{2}{3}(9, -9) = (2, 1)$. Since P is the midpoint of CQ , $Q = 2P - C = (1, -4)$. Thus $2x_Q - y_Q = 2 + 4 = 6$.

Question 18

A rectangular metal block measures $2\text{ cm} \times 3\text{ cm} \times 5\text{ cm}$ and has mass 270 g . What is its density in kg m^{-3} ?



Schematic block; dimensions are not drawn to scale.

A 90

B 900

C 9000

D 27000

E 90000

Worked solution

Answer C

The block has volume $2 \times 3 \times 5 = 30\text{ cm}^3 = 3 \times 10^{-5}\text{ m}^3$. Its mass is $270\text{ g} = 0.270\text{ kg}$.
Therefore its density is $\frac{0.270}{3 \times 10^{-5}} = 9000\text{ kg m}^{-3}$.

Question 19

Let O, A, B, C be the vertices of a parallelogram, with O as the origin and $\overrightarrow{OA} = \mathbf{a}$, $\overrightarrow{OB} = \mathbf{b}$ as adjacent sides. Thus C is the fourth vertex. Point P lies on AC with $AP : PC = 1 : 2$, and Q is the midpoint of OB . Which expression is $\overrightarrow{PQ}$?

A $\mathbf{a} + \frac{1}{6}\mathbf{b}$

B $-\mathbf{a} + \frac{1}{6}\mathbf{b}$

C $-\mathbf{a} - \frac{1}{6}\mathbf{b}$

D $\mathbf{a} - \frac{1}{6}\mathbf{b}$

E $-\mathbf{a} + \frac{1}{3}\mathbf{b}$

Worked solution

Answer B

Because OA and OB are adjacent sides, $\overrightarrow{OC} = \mathbf{a} + \mathbf{b}$, so $P = \mathbf{a} + \frac{1}{3}\mathbf{b}$. Also $Q = \frac{1}{2}\mathbf{b}$.
Therefore $\overrightarrow{PQ} = Q - P = -\mathbf{a} + \frac{1}{6}\mathbf{b}$.

Question 20

The line $y = 2x - 1$ is tangent to the graph $y = x^2 - 2kx + k + 3$, where k is real. What is the sum of the two possible values of k ?

A -3

B -2

C 3

D 1

E -1

Worked solution

Answer E

At an intersection, $x^2 - 2(k+1)x + k + 4 = 0$. Completing the square gives $(x - k - 1)^2 = k^2 + k - 3$. A tangent has exactly one point of contact, so the right-hand side must be zero.

Thus $k^2 + k - 3 = 0$, giving $k = \frac{-1+\sqrt{13}}{2}$ or $k = \frac{-1-\sqrt{13}}{2}$. The sum is -1 .

Tip: a repeated root can be recognised by completing the square; differentiation is not required.

Question 21

A quantity P is directly proportional to x^2 and inversely proportional to y . If x is increased by 20% and y is decreased by 20%, by what factor does P change?

A $\frac{4}{5}$

B $\frac{6}{5}$

C $\frac{9}{5}$

D 2

E $\frac{12}{5}$

Worked solution

Answer C

Write $P = c \frac{x^2}{y}$ for some constant c . After the changes, x is multiplied by $6/5$ and y by $4/5$, so P is multiplied by $\frac{(6/5)^2}{4/5} = \frac{9}{5}$.

Question 22

Six cards are numbered 1, 2, 3, 4, 5, 6. Two different cards are selected, with every pair equally likely.

Given that the product of their numbers is even, what is the probability that their sum is greater than 7?

A $\frac{1}{3}$

B $\frac{1}{5}$

C $\frac{5}{12}$

D $\frac{1}{2}$

E $\frac{3}{5}$

Worked solution

Answer C

There are $6 \times 5/2 = 15$ unordered pairs. The product is odd only when both cards are odd. The three odd cards form 3 pairs, so $15 - 3 = 12$ pairs satisfy the condition.

The eligible pairs whose sum exceeds 7 are

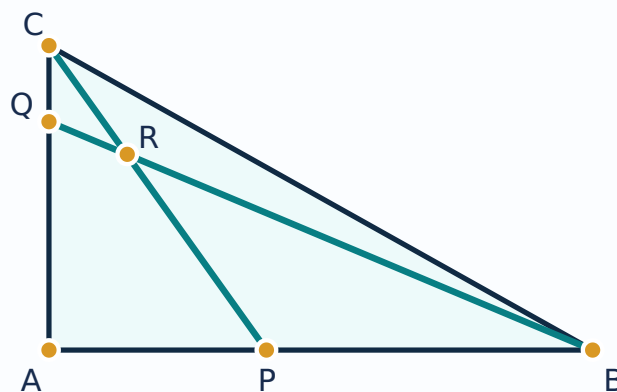
$$(2, 6), (3, 6), (4, 5), (4, 6), (5, 6).$$

The pair (3, 5) has a large enough sum but has an odd product. The required probability is $5/12$.

Useful approach: Apply the given condition to both the denominator and the favourable outcomes.

Question 23

In triangle ABC , let $A = (0, 0)$, $B = (10, 0)$, and $C = (0, 8)$. Point P lies on AB with $AP : PB = 2 : 3$, and point Q lies on AC with $AQ : QC = 3 : 1$. Lines CP and BQ meet at R . What is the x -coordinate of R ?



Schematic diagram; not to scale.

A 1

B $\frac{4}{3}$

C $\frac{5}{3}$

D $\frac{10}{7}$

E 2

Worked solution

Answer D

$P = (4, 0)$, $Q = (0, 6)$. The two lines are $CP : y = 8 - 2x$ and $BQ : y = 6 - \frac{3}{5}x$. Equating gives $2 = \frac{7}{5}x$, hence $x = 10/7$.

Question 24

Box A is chosen with probability $\frac{2}{3}$, and box B with probability $\frac{1}{3}$. Box A contains 2 red and 3 blue counters; box B contains 4 red and 1 blue counter. Two counters are drawn without replacement from the chosen box. Given that the two counters have different colours, what is the probability that box A was chosen?

A $\frac{1}{2}$

B $\frac{2}{3}$

C $\frac{7}{10}$

D $\frac{3}{4}$

E $\frac{4}{5}$

Worked solution

Answer D

Let D mean that the two drawn counters have different colours. From box A , the red-blue and blue-red branches give $P(D | A) = 2(\frac{2}{5})(\frac{3}{4}) = \frac{3}{5}$. For B , $P(D | B) = 2(\frac{4}{5})(\frac{1}{4}) = \frac{2}{5}$.

The probability of choosing A and then drawing different colours is $\frac{2}{3} \cdot \frac{3}{5} = \frac{2}{5}$. The probability of different colours overall is $\frac{2}{5} + \frac{1}{3} \cdot \frac{2}{5} = \frac{8}{15}$. Of these different-colour outcomes, the fraction originating from A is $\frac{2/5}{8/15} = \frac{3}{4}$.

Question 25

How many ordered pairs of positive integers (a, b) satisfy $ab = 360$, have highest common factor 1, and have $a + b$ a multiple of 7?

A 2

B 0

C 6

D 8

E 4

Worked solution

Answer E

Since $360 = 2^3 \cdot 3^2 \cdot 5$ and a, b are coprime, the prime powers 8, 9, 5 must be assigned wholly to one factor. The possible unordered splits are $1 + 360, 8 + 45, 9 + 40, 72 + 5$; only $9 + 40 = 49$ and $72 + 5 = 77$ are multiples of 7. Both orders count, giving 4.

Question 26

Data set A has mean 14, and data set B has mean 20. The mean of all observations in the two data sets combined is 16. An observation of value 10 from A is exchanged with an observation of value 34 from B . The two data sets then have the same mean. How many observations are there altogether?

A 6

B 12

C 18

D 24

E 36

Worked solution

Answer C

Exchanging the two observations changes neither the combined total nor the size of either data set. The combined mean remains 16, so the two equal new means must both be 16.

The total in A increases by $34 - 10 = 24$, while its mean increases by $16 - 14 = 2$. Therefore A contains $24/2 = 12$ observations. The total in B decreases by 24, while its mean decreases by $20 - 16 = 4$, so B contains $24/4 = 6$ observations. Altogether there are $12 + 6 = 18$.

Trap: averaging the two original means without accounting for the different data set sizes.

Question 27

Two geometrically similar solid objects are made from the same uniform material. Their total surface areas are in the ratio $4 : 9$, smaller to larger. The mass of the larger object is 95 g greater than the mass of the smaller object. What is the mass of the smaller object?

A 40 g

B 20 g

C 76 g

D 95 g

E 190 g

Worked solution

Answer A

The surface area ratio $4 : 9$ gives a length ratio $2 : 3$. The volume ratio is therefore $2^3 : 3^3 = 8 : 27$. Because the objects are made from the same uniform material, their masses also have ratio $8 : 27$.

The difference of $27 - 8 = 19$ ratio parts is 95 g, so one part is 5 g. The smaller mass is $8 \times 5 = 40$ g.

Trap: applying the surface area ratio directly to mass gives 76 g; using the length ratio gives 190 g.

MODULE 02

Mathematics 2

QUESTIONS & WORKED SOLUTIONS

Engineering pathway

27

questions

40

minutes

1

mark per correct answer

NO CALCULATOR

Using this module

Choose one answer per question. There is no negative marking.
Each worked solution appears immediately below its question.
Attempt the question first; cover the solution if working to time.
Use separate paper for your working.

Question 1 begins on the next page.

Question 01

The curve $y = \frac{(x+2)(x-1)}{\sqrt{x}}$ is defined for $x > 0$. What is the gradient of the normal to the curve at $x = 4$?

A $-\frac{27}{8}$

B $-\frac{8}{25}$

C $\frac{8}{27}$

D $-\frac{8}{27}$

E $\frac{27}{8}$

Worked solution

Answer D

Expand the numerator and divide each term by $\sqrt{x}$:

$$y = x^{3/2} + x^{1/2} - 2x^{-1/2}.$$

Differentiate the powers separately:

$$\frac{dy}{dx} = \frac{3}{2}x^{1/2} + \frac{1}{2}x^{-1/2} + x^{-3/2}.$$

At $x = 4$, the tangent gradient is $3 + \frac{1}{4} + \frac{1}{8} = \frac{27}{8}$. A normal is perpendicular to the tangent, so its gradient is the negative reciprocal, $-\frac{8}{27}$.

Question 02

A sequence is defined by $u_1 = 3$ and $u_{n+1} = \frac{3u_n - 1}{u_n + 1}$. What is $\prod_{n=1}^{20} u_n$?

A 21

B 42

C 120

D 220

E 231

Worked solution

Answer E

The first few terms are $3, 2, \frac{5}{3}$, suggesting $u_n = \frac{n+2}{n}$. This formula holds at $n = 1$. If it holds at a particular n , the recurrence gives

$$u_{n+1} = \frac{3(n+2)/n - 1}{(n+2)/n + 1} = \frac{2n+6}{2n+2} = \frac{n+3}{n+1},$$

so it holds at the next term as well. Consequently,

$$\prod_{n=1}^{20} u_n = \frac{3}{1} \cdot \frac{4}{2} \cdot \frac{5}{3} \cdots \frac{22}{20} = \frac{21 \cdot 22}{1 \cdot 2} = \boxed{231}.$$

All factors from 3 to 20 cancel between numerator and denominator.

Question 03

For $0 \leq x \leq 2\pi$, with the original expression defined, $\frac{\sin^2 x}{1 - \cos x} = 2 \sin x$. What is the sum of $\cos x$ over all distinct real solutions?

A $-\frac{2}{5}$

B $-\frac{1}{2}$

C 0

D $\frac{2}{5}$

E $\frac{1}{2}$

Worked solution

Answer A

The denominator excludes $x = 0, 2\pi$. For all allowed solutions, $1 + \cos x = 2 \sin x$, so $\sin x \geq 0$ and hence $0 < x \leq \pi$. Squaring and using $\sin^2 x = 1 - \cos^2 x$ gives $5 \cos^2 x + 2 \cos x - 3 = 0$. Thus $\cos x = 3/5$ or -1 , both satisfying the original equation. The sum is $-1 + 3/5 = -2/5$.

Question 04

A continuous function f satisfies $f(x) - f(-x) = 2x^3 - 6x$. What is $\int_{-2}^2 xf(x) dx$?

A $-\frac{16}{3}$

B $-\frac{16}{5}$

C $-\frac{4}{5}$

D $\frac{16}{5}$

E $\frac{16}{3}$

Worked solution

Answer B

Pair the integrand at x and $-x$: $xf(x) + (-x)f(-x) = x[f(x) - f(-x)] = 2x^4 - 6x^2$.
Therefore the integral is $\int_0^2 (2x^4 - 6x^2) dx = 64/5 - 16 = -16/5$.

Question 05

For a positive integer n , the coefficient of x^4 in $(1-x)^n(1+x)^{n+2}$ is 27. What is the coefficient of x^6 ?

A -48

B -36

C -24

D 24

E 48

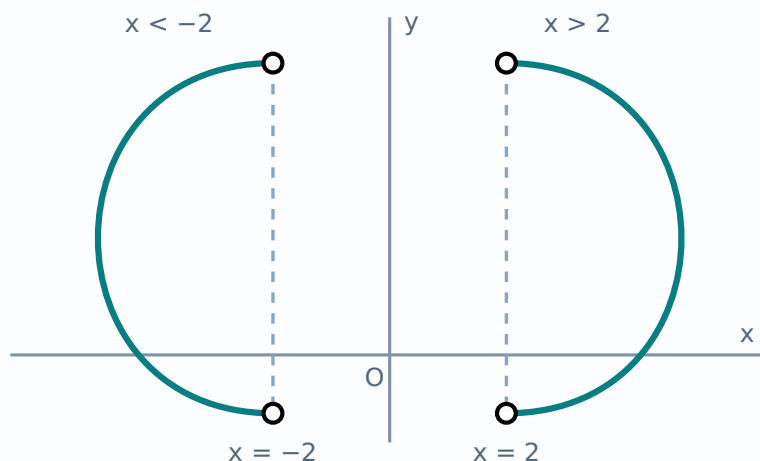
Worked solution

Answer A

Rewrite the product as $(1+x)^2(1-x^2)^n$. The coefficient of x^4 is $\binom{n}{2} - n = n(n-3)/2$, so $n = 9$. The coefficient of x^6 is $-\binom{9}{3} + \binom{9}{2} = -84 + 36 = -48$.

Question 06

The equation $\log_2(5 - y) + \log_2(y + 1) = 2\log_2(|x| - 2)$ defines a locus. What is the total length of the locus?



Open circles mark excluded endpoints; schematic, not to scale.

A $\frac{3\pi}{2}$

B 2π

C 4π

D 6π

E 9π

Worked solution

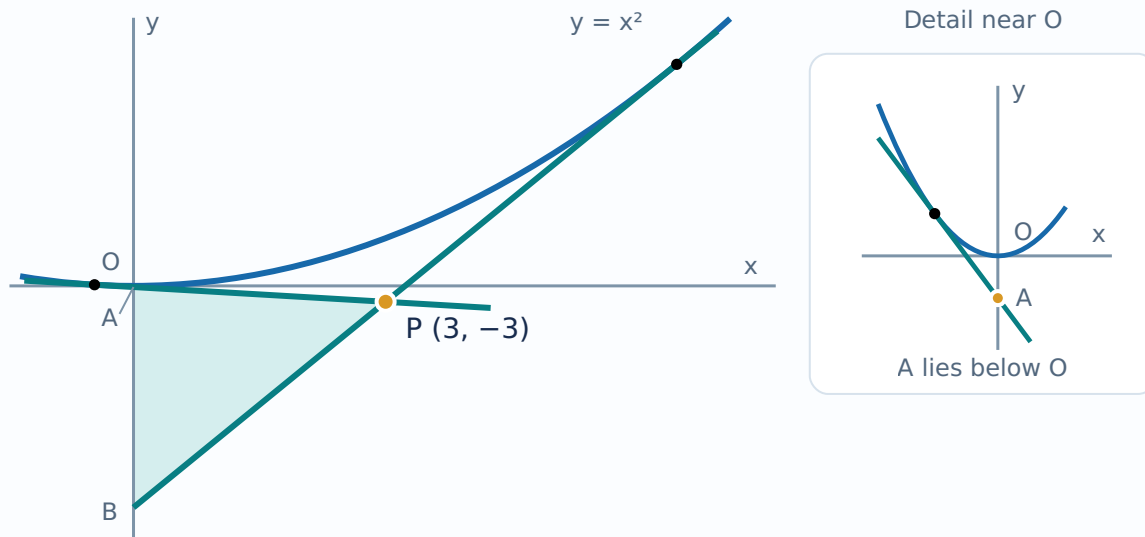
Answer D

The logarithms require $-1 < y < 5$ and $|x| > 2$. Combining and exponentiating gives $(y - 2)^2 + (|x| - 2)^2 = 9$. The restrictions select two open semicircular arcs, each of radius 3, so the total length is $2(3\pi) = 6\pi$.

Trap: keeping the forbidden halves of the two circles after completing the square.

Question 07

Two tangents are drawn from $(3, -3)$ to the parabola $y = x^2$. What is the area of the region enclosed by the two tangents and the y -axis?



The tangents and y -axis bound the shaded triangle. Axes use different scales; the inset enlarges the origin.

A $36\sqrt{3}$

B $18\sqrt{3}$

C $12\sqrt{3}$

D 36

E $72\sqrt{3}$

Worked solution

Answer A

A tangent through $(3, -3)$ with gradient m is $y + 3 = m(x - 3)$. Intersections with $y = x^2$ give $x^2 - mx + 3m + 3 = 0$; tangency gives $m^2 - 12m - 12 = 0$, so $m = 6 \pm 4\sqrt{3}$. The two y -axis intercepts differ by $3(8\sqrt{3}) = 24\sqrt{3}$. With horizontal distance 3 from $(3, -3)$ to the y -axis, the area is $\frac{1}{2}(24\sqrt{3})(3) = 36\sqrt{3}$.

Question 08

Let $f(x) = |x^2 - 4x + 3|$. For which positive value of k does $f(f(x)) = k$ have exactly 8 distinct real solutions?

A $\frac{3}{2}$

B $\frac{1}{2}$

C 1

D 2

E 3

Worked solution

Answer C

Write $f(x) = |(x - 2)^2 - 1|$. For $t \geq 0$, solve $f(x) = t$ by using $(x - 2)^2 = 1 + t$ or $(x - 2)^2 = 1 - t$. There are two solutions for $t = 0$, four for $0 < t < 1$, three for $t = 1$, and two for $t > 1$.

Now put $y = f(x)$, so $y \geq 0$. For $k = 1$, solving $f(y) = 1$ gives the three distinct nonnegative values

$$y = 2 - \sqrt{2}, \quad 2, \quad 2 + \sqrt{2}.$$

The first lies between 0 and 1, so it contributes four possible values of x . Each of the other two exceeds 1, so each contributes two. There are $4 + 2 + 2 = 8$ distinct solutions.

For the other listed values $k = \frac{3}{2}, \frac{1}{2}, 2, 3$, the same counting gives 6, 10, 6, 4 solutions respectively. Hence the required value is 1.

Question 09

A sequence has n th term $u_n = 6n - 1$. From the first 15 terms, every term whose position is a multiple of 3 is removed. What is the sum of the terms that remain?

A 265

B 435

C 440

D 450

E 705

Worked solution

Answer C

The first and fifteenth terms are 5 and 89, so the sum of all fifteen terms is

$$S = \frac{15}{2}(5 + 89) = 705.$$

The removed terms are $u_3, u_6, u_9, u_{12}, u_{15}$, namely 17, 35, 53, 71, 89. They form an arithmetic progression with sum

$$R = \frac{5}{2}(17 + 89) = 265.$$

The required sum is $705 - 265 = \boxed{440}$. The multiples of 3 refer to the term positions, not the values of the terms.

Question 10

For $0 < x < \frac{\pi}{2}$, $\sin^2 x + \sin^4 x + \sin^6 x + \cdots = 3 \cos^2 x$. What is $\tan^2 x$?

A $\frac{\sqrt{13} + 1}{2}$

B $\frac{\sqrt{13} - 1}{2}$

C $\frac{\sqrt{13} - 3}{2}$

D $\frac{\sqrt{13} + 3}{2}$

E 3

Worked solution

Answer B

The series is $\frac{\sin^2 x}{1 - \sin^2 x} = \tan^2 x$. Let $t = \tan^2 x$. Since $\cos^2 x = 1/(1 + t)$, $t = 3/(1 + t)$, so $t^2 + t - 3 = 0$. The positive root is $(\sqrt{13} - 1)/2$.

Question 11

For $0 \leq x \leq 4$, consider the region between the curve $y = x^2 - 4x + 3$ and the x -axis. What fraction of the total area of this region lies below the x -axis?

A $\frac{1}{4}$

B $\frac{1}{3}$

C $\frac{1}{2}$

D $\frac{2}{3}$

E $\frac{3}{4}$

Worked solution

Answer B

Factorise $x^2 - 4x + 3 = (x - 1)(x - 3)$. The curve is below the axis for $1 < x < 3$, and above it on the two remaining intervals.

An antiderivative is $F(x) = \frac{x^3}{3} - 2x^2 + 3x$. Its values at 0, 1, 3, 4 are $0, \frac{4}{3}, 0, \frac{4}{3}$, respectively. Thus each of the three separate regions has area $\frac{4}{3}$:

$$A_{\text{below}} = - \int_1^3 (x^2 - 4x + 3) dx = \frac{4}{3}, \quad A_{\text{total}} = 4.$$

The fraction below the axis is $\frac{4/3}{4} = \boxed{\frac{1}{3}}$. The signed integral over $[0, 4]$ is not the total area.

Question 12

The equation $3^{2x} - a3^x + 9 = 0$ has distinct real roots u, v satisfying $u^2 + v^2 = 10$. What is a ?

A $\frac{28}{3}$

B $\frac{244}{9}$

C 26

D $\frac{82}{3}$

E 28

Worked solution

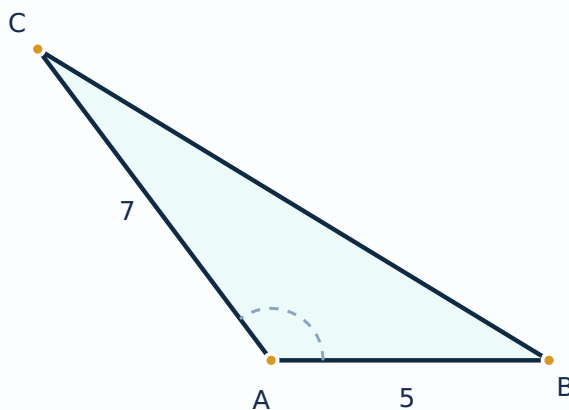
Answer D

Put $t = 3^x$. The quadratic in t has roots $3^u, 3^v$, whose product is 9, so $u + v = 2$. From $u^2 + v^2 = 10$, $uv = -3$, hence $\{u, v\} = \{-1, 3\}$. Therefore $a = 3^{-1} + 3^3 = 82/3$.

Trap: applying Vieta directly to the exponents instead of to $3^u, 3^v$.

Question 13

Triangle ABC has $AB = 5$, $AC = 7$, and area 14. Angle BAC is obtuse. What is the length of BC ?



Angle BAC is obtuse. No measurement from the diagram is required.

A $4\sqrt{2}$

B $\sqrt{46}$

C $\sqrt{74}$

D $\sqrt{102}$

E $2\sqrt{29}$

Worked solution

Answer E

Let $\theta = \angle BAC$. The area formula gives

$$14 = \frac{1}{2}(5)(7) \sin \theta, \quad \sin \theta = \frac{4}{5}.$$

Since the angle is obtuse, its cosine is negative. Using $\sin^2 \theta + \cos^2 \theta = 1$ gives $\cos \theta = -\frac{3}{5}$.

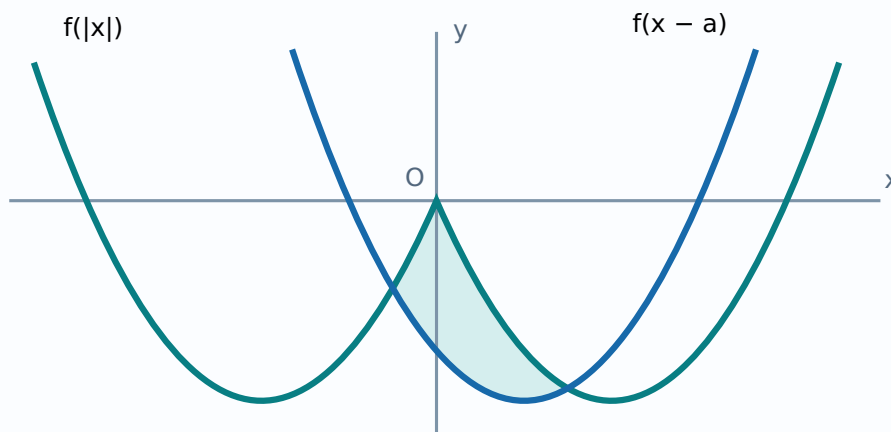
Now apply the cosine rule:

$$BC^2 = 5^2 + 7^2 - 2(5)(7) \left(-\frac{3}{5}\right) = 74 + 42 = 116.$$

Therefore $BC = \sqrt{116} = \boxed{2\sqrt{29}}$. Choosing the positive value of the cosine would give the acute-angle triangle instead.

Question 14

Let $f(x) = x^2 - 4x$ and suppose $-4 < a < 0$. The curves $y = f(|x|)$ and $y = f(x - a)$ enclose a region of area 3. What is the sum of the squares of all possible values of a ?



Representative configuration for $-4 < a < 0$; the shaded region lies between the curves.

A 6

B 16

C 12

D 8

E 10

Worked solution

Answer E

The difference $f(|x|) - f(x - a)$ is linear on each side of 0, with zeros at $x = a/2$ and $x = (a + 4)/2$. For $-4 < a < 0$, this difference is positive between its zeros. The region under the graph of the difference, above the x -axis, is a triangle of base 2 and height $-a(a + 4)$. Therefore the area between the original curves is $-a(a + 4)$. Setting this to 3 gives $a = -1, -3$, and the required sum of squares is $1 + 9 = 10$.

Question 15

A monic polynomial P of degree 4 has remainder $2x + 1$ when divided by $x^2 - 1$, and remainder $x + 7$ when divided by $x^2 - 4$. What is the least value of $P(x) + P(-x)$ for real x ?

A $\frac{3}{4}$

B 6

C $\frac{3}{2}$

D 3

E $\frac{9}{2}$

Worked solution

Answer C

Write $P(x) = x^4 + bx^3 + cx^2 + dx + e$. When reducing the polynomial using $x^2 = 1$, its constant part is $1 + c + e$. The first remainder therefore gives $1 + c + e = 1$, or $c + e = 0$.

Using $x^2 = 4$, the constant part is $16 + 4c + e$. The second remainder gives $16 + 4c + e = 7$, or $4c + e = -9$. Subtracting yields $c = -3$, and hence $e = 3$.

The odd powers cancel in the required sum:

$$P(x) + P(-x) = 2(x^4 - 3x^2 + 3) = 2\left(x^2 - \frac{3}{2}\right)^2 + \frac{3}{2}.$$

The squared term is zero at the real values $x = \pm\sqrt{\frac{3}{2}}$. Thus the least value is $\boxed{\frac{3}{2}}$.

Question 16

What is the total length of the intervals within $0 \leq x \leq 2\pi$ for which $2\sin^2 x + 3\cos x > 0$?

A $\frac{\pi}{3}$

B $\frac{2\pi}{3}$

C π

D $\frac{4\pi}{3}$

E $\frac{5\pi}{3}$

Worked solution

Answer D

Use $\sin^2 x = 1 - \cos^2 x$. The inequality becomes

$$2 + 3\cos x - 2\cos^2 x = (2 - \cos x)(2\cos x + 1) > 0.$$

For every real x , $-1 \leq \cos x \leq 1$, so $2 - \cos x$ is positive. Therefore the condition is simply $\cos x > -\frac{1}{2}$.

Within the stated domain, this holds on $[0, \frac{2\pi}{3})$ and $(\frac{4\pi}{3}, 2\pi]$. Their lengths add to

$\frac{2\pi}{3} + \frac{2\pi}{3} = \boxed{\frac{4\pi}{3}}$. Excluding isolated endpoints does not change an interval's length.

Question 17

A function f , defined for every real number, has exactly four distinct real zeros. The equation $f(x + 1/x) = 0$, with $x \neq 0$, has exactly five distinct real solutions. How many of the zeros of f must lie strictly between -2 and 2 ?

A 2

B 3

C 0

D 4

E 1

Worked solution

Answer E

For a zero r of f , the equation $x + 1/x = r$ becomes $x^2 - rx + 1 = 0$. Its discriminant is $r^2 - 4$, so this zero contributes zero, one, or two real solutions according as $|r| < 2$, $|r| = 2$, or $|r| > 2$. None of these quadratic roots is zero.

Let h be the number of zeros of f outside $[-2, 2]$, and let b be the number at the boundary values -2 and 2 . Different zeros give disjoint sets of solutions, so $2h + b = 5$, with $0 \leq b \leq 2$.

The left side is odd only when $b = 1$. Hence $h = 2$, leaving $4 - 2 - 1 = \boxed{1}$ zero strictly between -2 and 2 .

Question 18

Consider the claim: every real polynomial f with $f(0) < 0$, $f(2) > 0$, and $f'(0) > 0$ has exactly one zero in $(0, 2)$. Which polynomial is a counterexample?

A $-4x^3 + 12x^2 - 11x + 3$

B $x^3 + x - 1$

C $4x^3 - 12x^2 + 11x - 3$

D $x^3 - x - 1$

E $4x^3 - 12x^2 + 8x$

Worked solution

Answer C

Option C factorises as

$$4x^3 - 12x^2 + 11x - 3 = (2x - 1)(x - 1)(2x - 3).$$

It has three distinct zeros, $\frac{1}{2}$, 1 , $\frac{3}{2}$, all in $(0, 2)$. It also satisfies all three hypotheses: $f(0) = -3 < 0$, $f(2) = 3 > 0$, and $f'(0) = 11 > 0$. It is therefore a counterexample.

A and E fail the requirement $f(0) < 0$, while D fails $f'(0) > 0$. B does satisfy the hypotheses, but its derivative $3x^2 + 1$ is positive everywhere; it has exactly one zero in the interval, so it does not disprove the claim.

Question 19

Let n be a positive integer. Which condition is necessary and sufficient for n^3 to be divisible by 72 while n^2 is not divisible by 72?

A $n = 6k + 3$ for some integer $k \geq 0$

B n is divisible by 6

C $n = 12k + 6$ for some integer $k \geq 0$

D n is divisible by 12

E $n = 24k + 12$ for some integer $k \geq 0$

Worked solution

Answer C

Factorise $72 = 2^3 \cdot 3^2$. For 72 to divide n^3 , the integer n must contain at least one factor of 2 and at least one factor of 3. Equivalently, 6 divides n .

Then n^2 already contains the required 3^2 . It fails to be divisible by 72 precisely when it lacks 2^3 , which means that n contains exactly one factor of 2.

Thus n must be divisible by 6 but not by 12. This is exactly $n = 12k + 6$ for an integer $k \geq 0$. Every integer of this form also satisfies both original divisibility conditions.

Question 20

Let $f(x) = x^5 - 5ax^3 + 5x$. For which values of a is $f'(x) > 0$ for every real x ?

A $a \leq -\frac{2}{3}$

B $a < \frac{2}{3}$

C $a \leq \frac{2}{3}$

D $a > \frac{2}{3}$

E all real a

Worked solution

Answer B

Differentiate and put $t = x^2 \geq 0$:

$$f'(x) = 5(x^4 - 3ax^2 + 1) = 5(t^2 - 3at + 1).$$

If $a \leq 0$, all the terms in $t^2 - 3at + 1$ are nonnegative and the constant term is 1, so the expression is positive.

If $a > 0$, the quadratic has its minimum for $t \geq 0$ at $t = \frac{3a}{2}$. That minimum is $1 - \frac{9a^2}{4}$, which is strictly positive exactly when $a < \frac{2}{3}$.

Combining the cases gives $a < \frac{2}{3}$. At $a = \frac{2}{3}$, the derivative equals $5(x^2 - 1)^2$, which is zero at $x = \pm 1$, so equality is excluded.

Question 21

A differentiable function f satisfies $f'(x) = 6x - 12$ for every real x , and $\int_0^2 f(x) dx = 0$. What is $f(3)$?

A -9

B -1

C 1

D 7

E 15

Worked solution

Answer B

Integrating the derivative gives $f(x) = 3x^2 - 12x + C$, where C is a constant. Use the definite-integral condition to determine it:

$$0 = \int_0^2 (3x^2 - 12x + C) dx = \left[x^3 - 6x^2 + Cx \right]_0^2 = 8 - 24 + 2C.$$

Thus $C = 8$, and $f(3) = 27 - 36 + 8 = \boxed{-1}$. A zero definite integral does not mean that the function itself is zero at either endpoint.

Question 22

A real polynomial f of degree exactly 4 has exactly three distinct stationary points at $x = a, b, c$, where $a < b < c$. It satisfies $f(a) = f(c) = 0$ and $f(b) > 0$. Consider: I. $b = (a + c)/2$. II. $f(x) \geq 0$ for every real x . III. For every real $d > 0$, $f(x) = d$ has exactly four distinct real solutions. Which statements must be true?

A II only

B I and III only

C I only

D I, II and III

E I and II only

Worked solution

Answer E

A stationary point at which the polynomial equals zero is a repeated root: both f and f' vanish there. Thus a and c are each roots of multiplicity at least two. Since the degree is exactly four,

$$f(x) = K(x - a)^2(x - c)^2$$

for a nonzero constant K . Because $f(b) > 0$ and b differs from a, c , we have $K > 0$. Hence $f(x) \geq 0$ everywhere, proving II.

Expanding the two quadratic factors and differentiating gives the factorisation

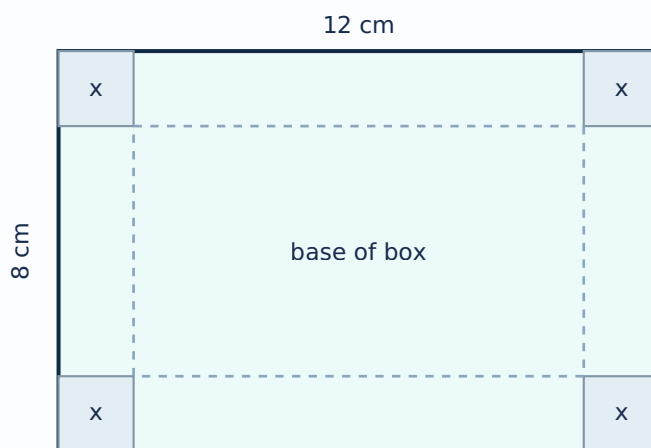
$$f'(x) = 2K(x - a)(x - c)(2x - a - c).$$

The third stationary point is therefore $b = (a + c)/2$, proving I. The graph has minima of zero at a, c and a maximum between them at b . For $d > f(b)$, only the two outer branches meet $y = d$, giving two solutions rather than four. III is false.

The correct selection is I and II only.

Question 23

A rectangular sheet measures 12 cm by 8 cm. A square of side x cm is cut from each corner, and the remaining sides are folded up to make an open box. For $0 < x < 4$, which value of x gives the greatest volume?



Shaded corner squares are removed. Dashed lines show the folds; x is illustrative.

A $\frac{10 - 2\sqrt{7}}{3}$

B $\frac{5 - \sqrt{7}}{3}$

C 2

D $\frac{10 - \sqrt{7}}{3}$

E $\frac{20 - 4\sqrt{7}}{3}$

Worked solution

Answer A

The box has height x , length $12 - 2x$, and width $8 - 2x$. Its volume is

$$V(x) = x(12 - 2x)(8 - 2x) = 96x - 40x^2 + 4x^3.$$

Differentiate and set the derivative equal to zero:

$$V'(x) = 96 - 80x + 12x^2 = 0 \iff 3x^2 - 20x + 24 = 0.$$

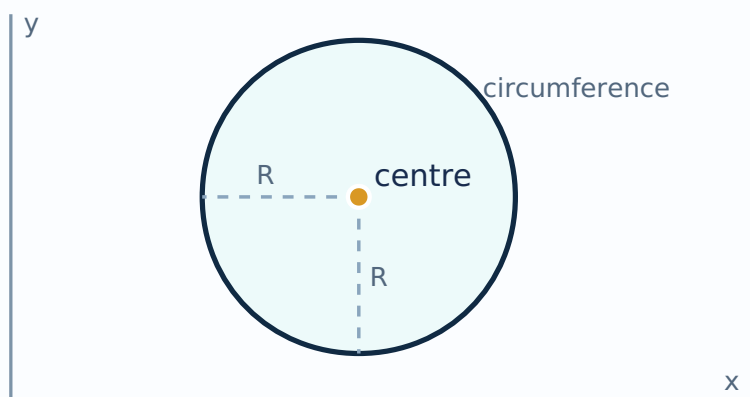
The roots are $x = \frac{10 \pm 2\sqrt{7}}{3}$. Since $2 < \sqrt{7} < 3$, the larger root exceeds 4, and only the smaller root lies in the permitted interval.

The upward-opening quadratic $V'(x)$ is positive before the smaller root and negative between the two roots. Hence the volume increases and then decreases on $(0, 4)$. Its

greatest value occurs at $x = \frac{10 - 2\sqrt{7}}{3}$.

Question 24

For real a, b, c , which condition is necessary and sufficient for $x^2 + y^2 + 2ax + 2by + c = 0$ to describe a nondegenerate circle whose entire circumference lies in the open first quadrant $x > 0, y > 0$?



Strict containment means the circle cannot touch either coordinate axis; schematic, not to scale.

A $a < 0, b < 0, \max(a^2, b^2) < c < a^2 + b^2$

B $a < 0, b < 0, \max(a^2, b^2) \leq c < a^2 + b^2$

C $a < 0, b < 0, 0 < c < a^2 + b^2$

D $a < 0, b < 0, c > a^2 + b^2$

E $a > 0, b > 0, \max(a^2, b^2) < c < a^2 + b^2$

Worked solution

Answer A

Completing the square gives centre $(-a, -b)$ and radius squared $R^2 = a^2 + b^2 - c$. Strict containment in the first quadrant requires $-a > R$ and $-b > R$, so $a, b < 0$, $c > a^2, b^2$, and $c < a^2 + b^2$ for a nondegenerate circle. This is option A.

Question 25

A geometric progression has first term $a > 0$ and real common ratio r . Both infinite series converge, and $\sum_{n=1}^{\infty} ar^{n-1} = 2$, while $\sum_{n=1}^{\infty} (ar^{n-1})^2 = 12$. What is $\sum_{n=1}^{\infty} |ar^{n-1}|$?

A 4

B 6

C 2

D 12

E 8

Worked solution

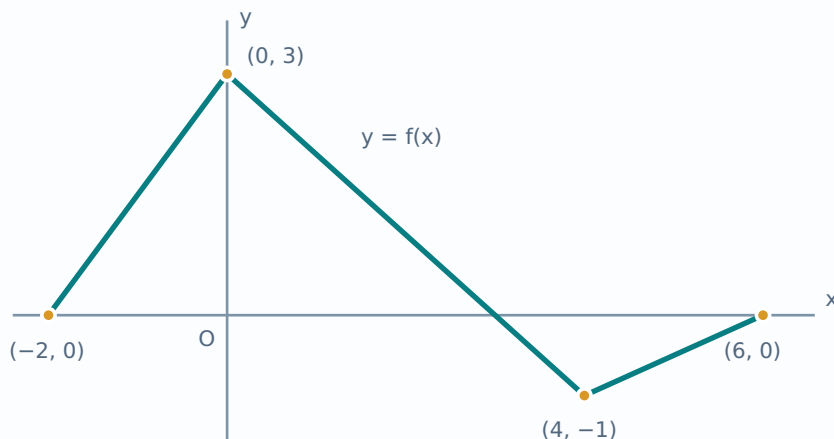
Answer B

Convergence gives $|r| < 1$, $a/(1-r) = 2$, and $a^2/(1-r^2) = 12$. Substitute $a = 2(1-r)$: $4(1-r)/(1+r) = 12$, so $r = -1/2$ and $a = 3$. The absolute values form a geometric series with first term 3, ratio $1/2$, and sum 6.

Trap: assuming r is positive; the signed sum forces a negative ratio.

Question 26

The function f is defined for $-2 \leq x \leq 6$ by the graph shown. For which values of x is $2f(3-x) + 1 < 0$?



The graph consists of straight line segments joining the labelled points.

A $-3 < x < -1$

B $-2 < x < -\frac{1}{2}$

C $-1 < x < \frac{1}{2}$

D $\frac{1}{2} < x < 2$

E $\frac{7}{2} < x < 5$

Worked solution

Answer B

First rearrange the inequality to $f(3-x) < -\frac{1}{2}$. Write $t = 3-x$ for the input to f .

On the descending segment from $(0, 3)$ to $(4, -1)$, the graph has equation $f(t) = 3 - t$, which reaches $-\frac{1}{2}$ at $t = \frac{7}{2}$. On the rising segment from $(4, -1)$ to $(6, 0)$, its equation is $f(t) = \frac{t-6}{2}$, which reaches $-\frac{1}{2}$ at $t = 5$. The graph is below this level precisely when

$$\frac{7}{2} < t < 5.$$

Substitute $t = 3-x$: $\frac{7}{2} < 3-x < 5$. Subtracting 3 and reversing the inequalities on multiplying by -1 gives $-2 < x < -\frac{1}{2}$. These inputs are within the domain of the transformed function.

Question 27

A geometric progression has positive terms. The sum of its first four terms is 30, and the sum of its first eight terms is 510. What is its sixth term?

A 32

B 48

C 60

D 64

E 128

Worked solution

Answer D

Let the first term be $a > 0$ and the common ratio be $r > 0$. Terms five to eight are r^4 times terms one to four. Their sum is $510 - 30 = 480$, so

$$30r^4 = 480, \quad r^4 = 16.$$

All terms are positive, so $r = 2$. The first four terms sum to $a(1 + 2 + 4 + 8) = 15a = 30$, giving $a = 2$. Therefore the sixth term is $ar^5 = 2 \cdot 2^5 = \boxed{64}$.

MODULE 03

Physics

QUESTIONS & WORKED SOLUTIONS

Engineering pathway

27

questions

40

minutes

1

mark per correct answer

NO CALCULATOR

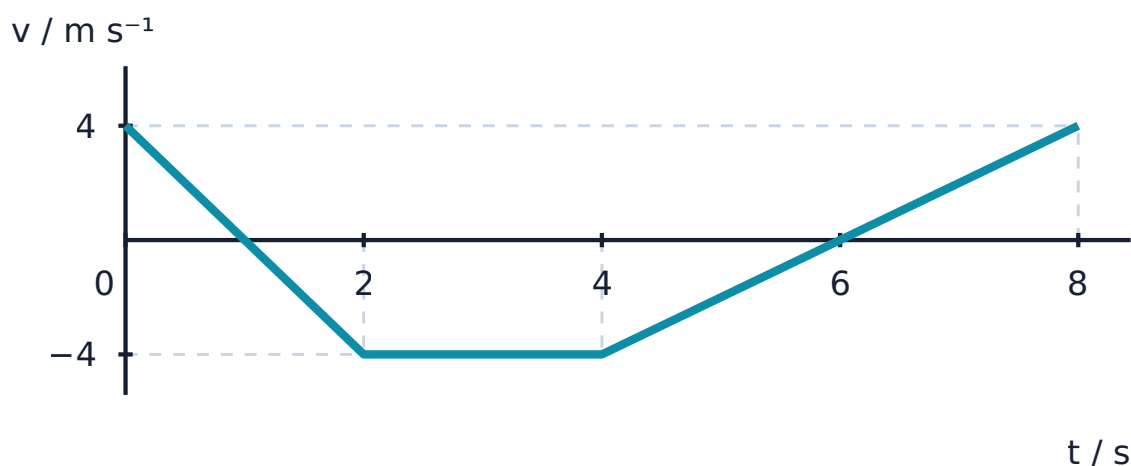
Using this module

Choose one answer per question. There is no negative marking.
Each worked solution appears immediately below its question.
Attempt the question first; cover the solution if working to time.
Use separate paper for your working.

Question 1 begins on the next page.

Question 01

The velocity of a particle varies with time as shown. What is the ratio of the total distance travelled to the magnitude of the displacement during the 8.0 s interval?



A 2

B $5/2$

C 3

D $7/2$

E 4

Worked solution

Answer B

From 0–2 s the signed area is zero, but the two triangular parts contribute 4 m of distance. From 2–4 s the particle has velocity -4 m s^{-1} , giving -8 m of displacement and 8 m of distance. From 4–8 s the signed area is again zero, while the two triangles contribute 8 m of distance. Thus the displacement is -8 m , its magnitude is 8 m, and the total distance is $4+8+8=20 \text{ m}$. The ratio is $20/8=5/2$.

Useful approach: Compute signed area and absolute area separately; a zero net area can still contain substantial distance.

Question 02

A solid cylinder A has radius r , height h and mass m . A second cylinder B is made from the same material, with radius $2r$ and height $h/2$. Both cylinders stand on circular bases. Cylinder B rests on a horizontal floor and A is placed centrally on B. What is the ratio of the pressure exerted on the floor by the stack to the pressure exerted by A alone on the floor?

A 1

B $1/4$

C $3/2$

D $1/2$

E $3/4$

Worked solution

Answer E

The volume of B is $\pi(2r)^2(h/2) = 2\pi r^2 h$, twice the volume of A, so B has mass $2m$. The stack therefore weighs $3mg$. The floor contact area is B's base area, $4\pi r^2$, so the stack pressure is $3mg/(4\pi r^2)$. A alone would exert $mg/(\pi r^2)$. Their ratio is $3/4$.

Useful approach: Find both the new weight and the new floor contact area before taking the ratio.

Question 03

An ideal battery supplies a network consisting of one resistor R in series with two resistors R in parallel. This network is replaced by two branches in parallel, each branch containing two resistors R in series. The battery voltage is unchanged. What is the ratio of the new total power to the original total power?

A $1/2$

B $2/3$

C 1

D $3/2$

E 2

Worked solution

Answer D

Initially, $R_{\text{eq}} = R + (R/2) = 3R/2$. In the replacement network each branch is $2R$, so the two branches have equivalent resistance R . At fixed voltage, total power is proportional to $1/R_{\text{eq}}$. Hence $P_{\text{new}}/P_{\text{old}} = (1/R)/(2/(3R)) = 3/2$.

Useful approach: Reduce the network before using $P = V^2/R$; the battery voltage is the fixed quantity.

Question 04

A spring stores 0.18 J when its extension is 3.0 cm. It obeys Hooke's law for all extensions up to 5.0 cm. What is the average magnitude of the spring force while it is extended at constant speed from 3.0 cm to 5.0 cm?

A 8 N

B 12 N

C 16 N

D 20 N

E 32 N

Worked solution

Answer C

At 3.0 cm, $0.18 = \frac{1}{2}kx^2$, so $k = 2 \times 0.18 / 0.030^2 = 400 \text{ N m}^{-1}$. The spring energies at 3.0 cm and 5.0 cm are therefore 0.18 J and $\frac{1}{2} \times 400 \times 0.050^2 = 0.50 \text{ J}$. The work done in the interval is 0.32 J over 0.020 m, so the average force is $0.32 / 0.020 = 16 \text{ N}$. Equivalently, it is the mean of the endpoint forces 12 N and 20 N.

Useful approach: For a linear force-extension relation, average force over an interval is the mean of the endpoint forces.

Question 05

A detector has a constant background count of 12 counts per minute. The total count from a radioactive source is 172 counts per minute initially and 92 counts per minute 20 minutes later. How many minutes after the initial reading will the total recorded count rate be 32 counts per minute? Treat the readings as mean count rates.

A 20 min

B 30 min

C 40 min

D 60 min

E 80 min

Worked solution

Answer D

Subtracting the background gives source counts of 160 initially and 80 after 20 minutes, so the half-life is 20 minutes. A total count of 32 means a source count of $32 - 12 = 20$. This is $160 \rightarrow 80 \rightarrow 40 \rightarrow 20$, three half-lives, so the time is $3 \times 20 = 60$ min.

Useful approach: Translate the requested total count back into a source count before counting half-lives.

Question 06

A vehicle travels one quarter of the total journey distance at 12 m s^{-1} , one half at 18 m s^{-1} and the final quarter at 36 m s^{-1} . What is its average speed for the whole journey?

A 18 m s^{-1}

B 15 m s^{-1}

C 20 m s^{-1}

D 12 m s^{-1}

E 24 m s^{-1}

Worked solution

Answer A

Let the journey length be D . The total time is $D/(4 \times 12) + D/(2 \times 18) + D/(4 \times 36) = D/48 + D/36 + D/144 = 8D/144 = D/18$. Thus the average speed is $D/(D/18) = 18 \text{ m s}^{-1}$.

Useful approach: Set a convenient total distance, such as 144 m, if fractions feel awkward.

Question 07

A 0.20 kg trolley moving at 6.0 m s^{-1} collides with a stationary 0.40 kg trolley and the two stick together. What fraction of the initial kinetic energy remains immediately after the collision?

A $1/9$

B $1/3$

C $1/2$

D $2/3$

E $3/4$

Worked solution

Answer B

Momentum gives $0.20 \times 6 = 0.60v$, so $v = 2 \text{ m s}^{-1}$. Initial kinetic energy is 3.6 J; final kinetic energy is 1.2 J. The fraction remaining is $1.2/3.6 = 1/3$.

Useful approach: Use momentum during the collision, then compare kinetic energies; sticking does not conserve kinetic energy.

Question 08

Car A has mass m , speed v and is stopped by a constant braking force F . Car B has mass $2m$, speed $2v$ and is stopped by a constant braking force $2F$. What is the ratio of B's stopping distance to A's?

A $1/2$

B 1

C 2

D 4

E 8

Worked solution

Answer D

Stopping distance is proportional to kinetic energy divided by braking force. B has $\frac{1}{2}(2m)(2v)^2 = 8 \times \frac{1}{2}mv^2$, while its braking force is only twice as large. Therefore its distance is $8/2 = 4$ times A's.

Useful approach: Write the proportionality $d \propto mv^2/F$ before inserting the scale factors.

Question 09

In a wind tunnel, the resistive forces on two similarly shaped falling objects are measured.

Speed / m s^{-1}	2	4	6
Object X drag / N	1	4	9
Object Y drag / N	2	8	18

Object X weighs 9 N and object Y weighs 18 N. Ignoring upthrust, which statement is correct about their terminal velocities in still air?

- A** Both objects reach terminal velocity at 6 m s^{-1} .
- B** Y has the greater terminal speed because it is heavier.
- C** X reaches terminal velocity at 4 m s^{-1} and Y at 6 m s^{-1} .
- D** Neither object reaches terminal velocity in the range shown.
- E** X reaches terminal velocity at 6 m s^{-1} and Y at 4 m s^{-1} .

Worked solution

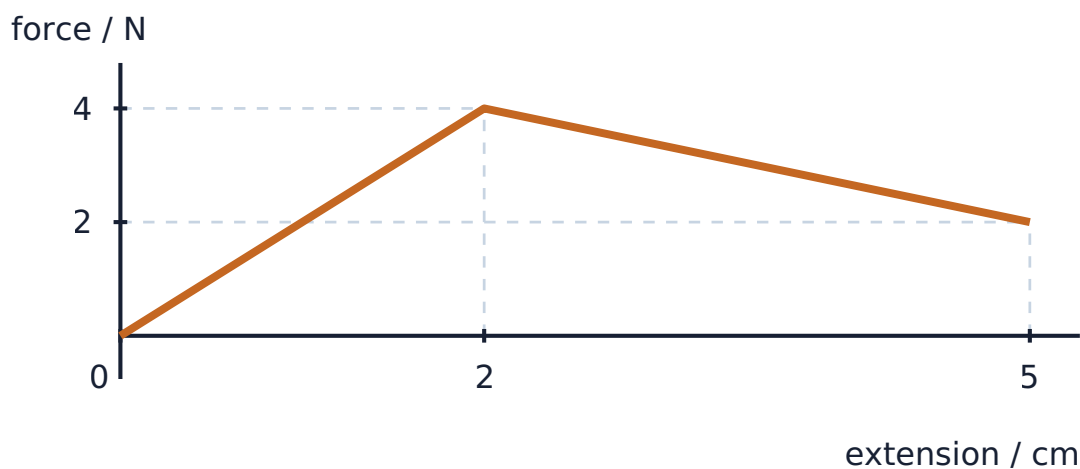
Answer A

Terminal velocity occurs when drag equals weight. At 6 m s^{-1} , X has drag 9 N, equal to its weight, and Y has drag 18 N, equal to its weight. Both therefore reach terminal velocity at 6 m s^{-1} .

Useful approach: Do not infer terminal velocity from mass alone: compare the resistive force with the weight.

Question 10

The graph shows the force applied to a wire as it is stretched slowly. What is the work done on the wire in increasing its extension from zero to 5.0 cm?



A 0.09 J

B 0.16 J

C 0.11 J

D 0.20 J

E 0.13 J

Worked solution

Answer E

The work done is the area under the loading force-extension graph. The first triangle has area $\frac{1}{2} \times 4 \times 0.020 = 0.040$ J. The second section is a trapezium with area $\frac{1}{2}(4+2) \times 0.030 = 0.090$ J. Total work done is 0.130 J.

This loading graph alone does not determine the recoverable elastic energy; that would also require information about unloading.

Useful approach: Convert centimetres to metres before multiplying force by extension.

Question 11

An NTC thermistor and a fixed resistor are connected in series across an ideal battery. A voltmeter is connected across the thermistor. The temperature of the thermistor decreases. Which change occurs?

- A** Thermistor resistance decreases, current increases, voltmeter reading increases.
- B** Thermistor resistance increases, current increases, voltmeter reading decreases.
- C** Thermistor resistance decreases, current decreases, voltmeter reading decreases.
- D** Thermistor resistance and current are unchanged, but the voltmeter reading increases.
- E** Thermistor resistance increases, current decreases, voltmeter reading increases.

Worked solution

Answer E

An NTC thermistor has greater resistance at lower temperature. The total series resistance therefore increases, so the current decreases. The thermistor's share of the fixed battery voltage increases because its resistance is now a larger fraction of the total. The voltmeter reading increases.

Useful approach: Track the chain: NTC cooler → resistance rises → total current falls → larger fraction of the fixed voltage lies across the thermistor.

Question 12

Water waves travel at 12 cm s^{-1} in two trials in a ripple tank. The water depth is unchanged. In each trial, straight wavefronts approach a gap in a barrier.

Trial	Frequency / Hz	Gap width / cm
P	4.0	6.0
Q	2.0	3.0

Compared with trial P, how do the wavelength and the spreading of the waves after the gap change in trial Q?

- A** The wavelength is twice as large; there is less spreading.
- B** The wavelength is twice as large; there is more spreading.
- C** The wavelength is half as large; there is more spreading.
- D** The wavelength is half as large; there is less spreading.
- E** The wavelength is unchanged; there is more spreading.

Worked solution

Answer B

Using $v = f\lambda$, the wavelengths are $12/4 = 3.0 \text{ cm}$ in P and $12/2 = 6.0 \text{ cm}$ in Q. The wavelength doubles.

Diffraction produces greater spreading when the wavelength is larger relative to the gap width. The wavelength-to-gap ratios are $3/6 = 1/2$ in P and $6/3 = 2$ in Q, so Q produces more spreading.

Useful approach: Compare wavelength with gap width; the absolute gap width alone does not determine diffraction.

Question 13

A fixed electrical power is transmitted through cables whose resistance does not change. The transmission voltage is increased by a factor of 4. By what factor does the power wasted in the cables change?

A It is unchanged.

B It becomes 1/2 as large.

C It becomes 1/4 as large.

D It becomes 1/8 as large.

E It becomes 1/16 as large.

Worked solution

Answer E

Since transmitted power is fixed and $P=VI$, increasing voltage fourfold makes current one quarter as large. Cable loss is I^2R , so the loss becomes $(1/4)^2 = 1/16$ as large.

Useful approach: Apply the voltage change to current first, then square the current for cable loss.

Question 14

An ideal transformer has 600 turns on its primary coil and 120 turns on its secondary coil. The secondary supplies a 24 V, 2.0 A load. What are the primary voltage and primary current?

A 24 V and 2.0 A

B 120 V and 0.40 A

C 120 V and 10 A

D 4.8 V and 10 A

E 600 V and 0.08 A

Worked solution

Answer B

The turns ratio is $N_p/N_s = 600/120 = 5$, so the primary voltage is $5 \times 24 = 120$ V. For an ideal transformer, input and output powers are equal: $V_p I_p = 24 \times 2$. Hence $I_p = 48/120 = 0.40$ A.

Useful approach: Use the turns ratio for voltage, then power conservation for current.

Question 15

10 g of ice at 0 °C is added to 100 g of water at 30 °C in an insulated container of negligible heat capacity. Use $c = 4.0 \text{ J g}^{-1} \text{ }^{\circ}\text{C}^{-1}$ and latent heat of fusion $L = 320 \text{ J g}^{-1}$. What is the final temperature?

A 0 °C

B 10 °C

C 20 °C

D 25 °C

E 30 °C

Worked solution

Answer C

Cooling the water to 0 °C releases $100 \times 4 \times 30 = 12000 \text{ J}$. Melting the ice uses $10 \times 320 = 3200 \text{ J}$, leaving 8800 J to warm 110 g of water. The temperature rise is $8800 / (110 \times 4) = 20 \text{ }^{\circ}\text{C}$, so the final temperature is 20 °C.

Useful approach: Check the phase-change stage first; only leftover energy heats the final mixture.

Question 16

A fixed amount of gas initially has pressure p , volume V and absolute temperature T . It is compressed at constant temperature to volume $V/2$. It is then heated at constant volume until its pressure is $3p$. For this gas at constant volume, pressure is directly proportional to absolute temperature. What is its final absolute temperature in terms of T ?

A $T/2$

B $2T/3$

C T

D $3T/2$

E $2T$

Worked solution

Answer D

After the isothermal compression, pressure doubles to $2p$. At constant volume, pressure is proportional to absolute temperature. Increasing pressure from $2p$ to $3p$ therefore multiplies temperature by $3/2$, giving $3T/2$.

Useful approach: Record the intermediate pressure before applying the second process.

Question 17

A light wave has wavelength λ in air. It enters a transparent material where its speed is $3c/4$, compared with speed c in air. How many complete oscillations occur while the wave travels normally through the parallel faces of a slab of thickness 6λ inside the material?

A 4

B 6

C 8

D 10

E 12

Worked solution

Answer C

Frequency remains unchanged at the boundary. Therefore the material wavelength is $3\lambda/4$. The number of wavelengths in the slab is $6\lambda/(3\lambda/4) = 8$, so eight complete oscillations occur.

Useful approach: Find the new wavelength before dividing the slab thickness by it.

Question 18

A source is tested using a detector in a fixed position. The background is 20 counts per minute. The table gives mean total count rates.

Absorber between source and detector	Total count rate / min^{-1}
None	500
Thin paper	260
Aluminium sheet	80

For this experiment, paper absorbs all alpha radiation and has negligible effect on beta and gamma radiation. The aluminium absorbs all alpha and beta radiation and has negligible effect on gamma radiation.

What fraction of the detected source count rate with no absorber is due to alpha and gamma radiation combined?

A $\frac{3}{8}$

B $\frac{1}{2}$

C $\frac{5}{8}$

D $\frac{3}{4}$

E $\frac{4}{5}$

Worked solution

Answer C

The source count rate without an absorber is $500 - 20 = 480 \text{ min}^{-1}$.

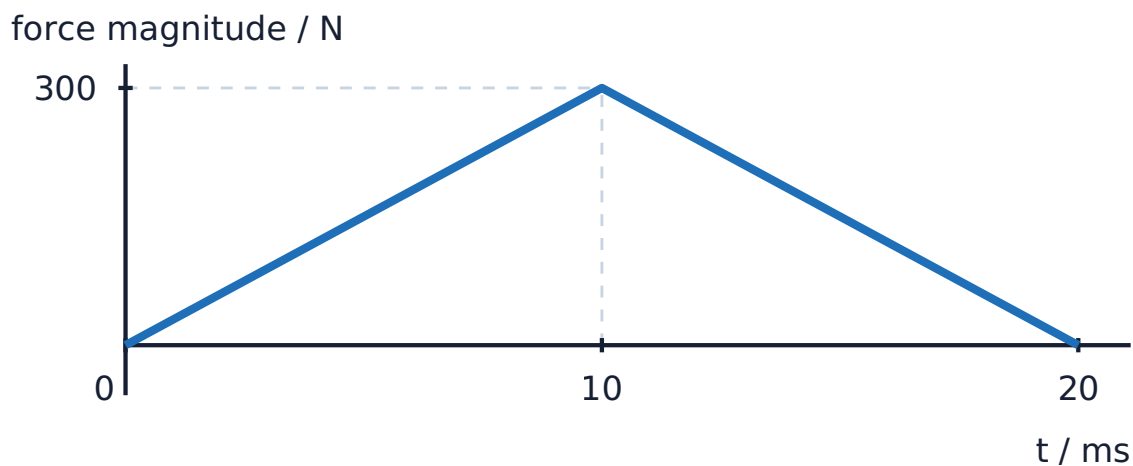
The drop caused by paper is the detected alpha contribution: $500 - 260 = 240 \text{ min}^{-1}$. The count remaining behind aluminium is gamma plus background, so the detected gamma contribution is $80 - 20 = 60 \text{ min}^{-1}$.

The required fraction is $(240 + 60)/480 = 300/480 = 5/8$. The beta contribution, $260 - 80 = 180 \text{ min}^{-1}$, accounts for the remaining $3/8$.

Useful approach: Subtract the background from a remaining total count, but not a second time from the difference between two total counts.

Question 19

A 0.20 kg ball travels to the right at 10 m s^{-1} and rebounds from a stationary wall. The force exerted by the wall on the ball is to the left and has the magnitude shown in the triangular force-time graph. Neglect any other horizontal impulse during the collision. What is the speed of the ball immediately after the collision?



A 5 m s^{-1}

B 10 m s^{-1}

C 15 m s^{-1}

D 20 m s^{-1}

E 30 m s^{-1}

Worked solution

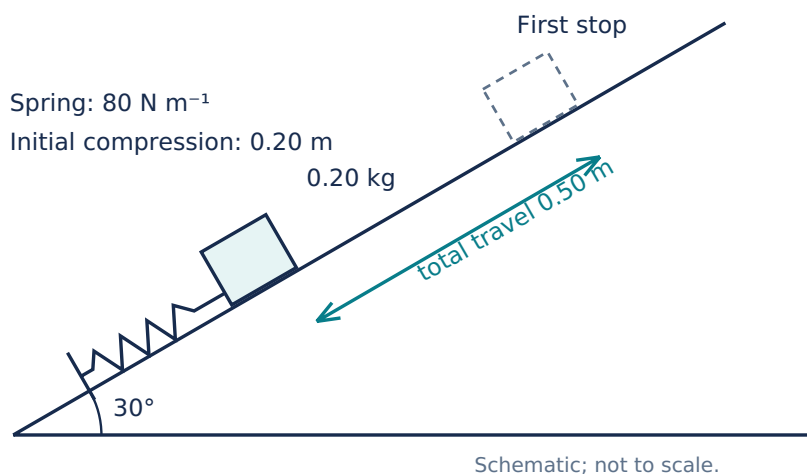
Answer A

The impulse is the triangular area $\frac{1}{2} \times 300 \times 0.020 = 3 \text{ N s}$ to the left. Taking right as positive, the initial momentum is $0.20 \times 10 = +2 \text{ N s}$ and the final momentum is $2 - 3 = -1 \text{ N s}$. The final velocity is $-1/0.20 = -5 \text{ m s}^{-1}$, so the rebound speed is **5 m s^{-1}** .

Useful approach: Convert milliseconds to seconds and assign a sign before applying $\Delta p =$ impulse.

Question 20

A 0.20 kg block is initially at rest on a rough 30° slope, against a light spring of spring constant 80 N m^{-1} compressed by 0.20 m along the slope. The block is released, moves up the slope and loses contact with the spring when it reaches its natural length. It first comes to rest after travelling a total of 0.50 m along the slope, measured from its initial compressed position. The friction force has constant magnitude; neglect other energy losses. Taking $g = 10 \text{ N kg}^{-1}$, what is the friction force on the block?



A 1.1 N

B 1.6 N

C 2.2 N

D 3.2 N

E 4.4 N

Worked solution

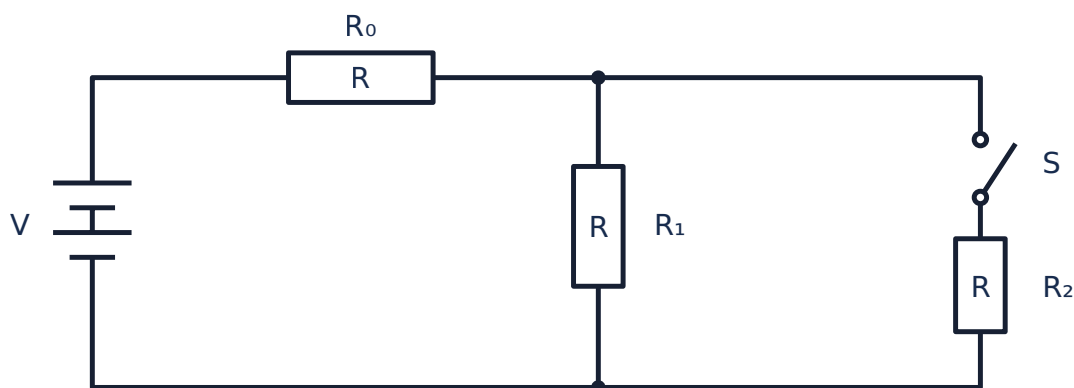
Answer C

The spring supplies $\frac{1}{2}kx^2 = \frac{1}{2} \times 80 \times 0.20^2 = 1.6 \text{ J}$. The vertical rise is $0.50 \sin 30^\circ = 0.25 \text{ m}$, so gravitational potential energy increases by $0.20 \times 10 \times 0.25 = 0.50 \text{ J}$. Friction therefore removes 1.10 J. Since its distance is 0.50 m, the friction force is $1.10/0.50 = 2.2 \text{ N}$.

Useful approach: Use spring energy = gravitational gain + work against friction.

Question 21

An ideal battery of fixed voltage V supplies the circuit shown. Each resistor has resistance R . Switch S is initially closed and is then opened. What is the ratio of the new power dissipated in R_1 to its original power?



Switch S shown open.

A $4/9$

B $3/4$

C 1

D $3/2$

E $9/4$

Worked solution

Answer E

With S closed, the parallel pair has resistance $R/2$, so the total resistance is $3R/2$. The voltage across R_1 is one third of the battery voltage: $V/3$. Its power is therefore $P_{\text{old}} = (V/3)^2/R = V^2/(9R)$.

With S open, R_0 and R_1 are in series and share the voltage equally. The voltage across R_1 becomes $V/2$, giving $P_{\text{new}} = V^2/(4R)$. Hence $P_{\text{new}}/P_{\text{old}} = 9/4$.

Useful approach: Track the voltage across the named resistor. Its power increases even though the total battery power decreases.

Question 22

A simple AC generator rotates at constant speed in a uniform magnetic field. At position A, the active sides of the coil move parallel to the field. After a quarter-turn, at position B, they move perpendicular to the field. Which statement is correct? Any change in rotation frequency leaves the coil and magnetic field unchanged.

- A** The voltage is zero at A and maximum in magnitude at B; doubling rotation frequency doubles both peak voltage and output frequency.
- B** The voltage is maximum in magnitude at A and zero at B; doubling rotation frequency doubles both peak voltage and output frequency.
- C** The voltage is zero at A and maximum in magnitude at B; doubling rotation frequency leaves peak voltage unchanged and doubles output frequency.
- D** The voltage is zero at A and maximum in magnitude at B; doubling rotation frequency doubles peak voltage and halves output frequency.
- E** The voltage is zero at A and maximum in magnitude at B; doubling rotation frequency halves peak voltage and doubles output frequency.

Worked solution

Answer A

Motion parallel to the field cuts no field lines, so the induced voltage is zero at A. Perpendicular motion gives the greatest rate of cutting, so the voltage has maximum magnitude at B. Doubling the rotation frequency doubles both the peak voltage and the output frequency.

Useful approach: Use the direction and speed of motion across the magnetic field to determine the induced voltage.

Question 23

Two slabs A and B have the same cross-sectional area and thickness. They are in perfect thermal contact, with their sides insulated. The outer face of A is held at $100\text{ }^{\circ}\text{C}$ and the outer face of B at $20\text{ }^{\circ}\text{C}$. Their thermal conductivities are k and $3k$, respectively.

For a fixed area and thickness, the rate of heat transfer through a slab is proportional to its thermal conductivity multiplied by the temperature difference across it. What is the temperature at the interface in steady state?

A $40\text{ }^{\circ}\text{C}$

B $30\text{ }^{\circ}\text{C}$

C $60\text{ }^{\circ}\text{C}$

D $80\text{ }^{\circ}\text{C}$

E $20\text{ }^{\circ}\text{C}$

Worked solution

Answer A

In steady state, the same rate of heat transfer passes through both slabs. If the interface temperature is $T\text{ }^{\circ}\text{C}$, the supplied relationship gives $k(100-T) = 3k(T-20)$. Cancelling k gives $100-T = 3T-60$, so $4T = 160$ and **$T = 40\text{ }^{\circ}\text{C}$** .

Equivalently, the less conductive slab A has three times the temperature drop of B: $60\text{ }^{\circ}\text{C}$ across A and $20\text{ }^{\circ}\text{C}$ across B.

Useful approach: The less conductive slab has the larger temperature drop.

Question 24

An ultrasound probe is placed on the front surface of a metal plate of thickness 0.30 m. Pulses travel normally to the plate faces. Measured from emission, an echo from the far surface returns after 0.10 ms and an echo from a crack returns after 0.020 ms. Assuming the ultrasound speed is the same on both paths, how far below the front surface is the crack?

A 0.03 m

B 0.06 m

C 0.10 m

D 0.12 m

E 0.24 m

Worked solution

Answer B

The far-surface echo travels 0.60 m in 0.10 ms, giving speed $0.60/0.00010 = 6000 \text{ m s}^{-1}$. The crack echo travels $6000 \times 0.000020 = 0.12 \text{ m}$ in total, so its depth is half of that: **0.06 m**.

Alternatively, both times are for return journeys at the same speed, so the depth ratio equals the time ratio: $0.30 \times (0.020/0.10) = 0.06 \text{ m}$.

Useful approach: Every echo time represents a journey to the reflector and back.

Question 25

A lamp has the following current–voltage readings:

Voltage across lamp	Current
2 V	0.30 A
4 V	0.50 A
6 V	0.60 A

The lamp is connected to a 12 V ideal supply through a $20\ \Omega$ resistor. Assuming the lamp current increases continuously with voltage throughout the operating range, which interval must contain the circuit current?

A Less than 0.20 A

B Between 0.20 A and 0.30 A

C Between 0.30 A and 0.50 A

D Between 0.50 A and 0.60 A

E Greater than 0.60 A

Worked solution

Answer C

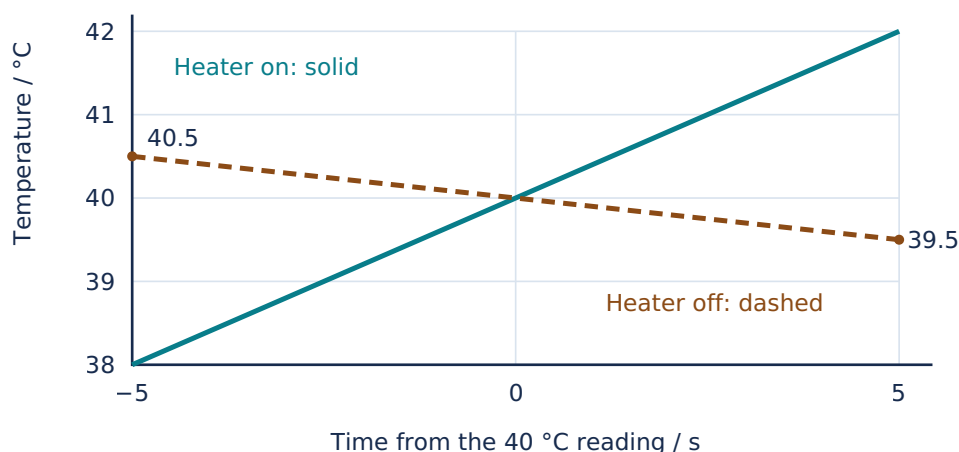
At 0.30 A, the resistor and lamp together would require $20 \times 0.30 + 2 = 8\text{ V}$, below 12 V, so current must be higher. At 0.50 A they would require $20 \times 0.50 + 4 = 14\text{ V}$, above 12 V, so current must be lower. It lies between 0.30 A and 0.50 A.

Useful approach: Test candidate currents in the total-voltage equation; do not replace the lamp with a fixed resistance.

Question 26

A metal block of mass 0.30 kg is heated by a 60 W heater. A second measurement is made while the same block cools with the heater off.

The straight lines below are tangents to the two temperature–time curves at 40°C . The rate of heat loss at this temperature is the same in both measurements. Neglect the heat capacities of the heater and thermometer, and assume the block has a uniform temperature.



What is the specific heat capacity of the metal?

A $200 \text{ J kg}^{-1} ^\circ\text{C}^{-1}$

B $400 \text{ J kg}^{-1} ^\circ\text{C}^{-1}$

C $500 \text{ J kg}^{-1} ^\circ\text{C}^{-1}$

D $667 \text{ J kg}^{-1} ^\circ\text{C}^{-1}$

E $2000 \text{ J kg}^{-1} ^\circ\text{C}^{-1}$

Worked solution

Answer B

The heating gradient is $(42 - 38)/10 = 0.40^\circ\text{C s}^{-1}$. The magnitude of the cooling gradient is $(40.5 - 39.5)/10 = 0.10^\circ\text{C s}^{-1}$.

Let H be the rate of heat loss at 40°C , and let c be the specific heat capacity. During cooling, $H = mc(0.10)$. With the heater on, $60 - H = mc(0.40)$.

Adding these energy-rate equations gives $60 = mc(0.50)$, so

$$c = \frac{60}{0.30 \times 0.50} = 400 \text{ J kg}^{-1} ^\circ\text{C}^{-1}.$$

Useful approach: At the same temperature, add the magnitudes of the heating and cooling slopes to correct for heat loss.

Question 27

A cart of mass m is released from rest and descends a frictionless track through a vertical height h . At the bottom it sticks to a stationary cart of mass $2m$. The combined carts then rise frictionlessly. What maximum vertical height do they reach above the bottom?

A $h/18$

B $h/9$

C $h/3$

D $h/2$

E $2h/3$

Worked solution

Answer B

Before the collision, $mgh = \frac{1}{2}mv^2$, so $v^2 = 2gh$. Momentum during the collision gives $mv = 3mu$, so $u = v/3$. After the collision, $\frac{1}{2}(3m)u^2 = (3m)gH$, hence $H = u^2/(2g) = h/9$.

Useful approach: Do not conserve kinetic energy through the sticking collision; use momentum only at that instant.